Synoptic Question 20: The Physics of a Nuclear Submarine

Reading Passage

Read the following article and then answer the questions that follow.

A nuclear-powered submarine is a complex feat of engineering, capable of operating submerged for months at a time. Its "engine" is a compact nuclear fission reactor, which generates an immense amount of thermal energy. This heat is used in a heat exchanger to boil water, creating high-pressure steam that spins turbines. These turbines, in turn, drive generators to produce electricity for the vessel's systems and main electric motors for propulsion.

The submarine's ability to submerge and surface is controlled by ballast tanks. To submerge, these tanks are flooded with seawater, increasing the submarine's overall mass. To surface, compressed air is used to force the water out, decreasing the mass. The submarine is in neutral buoyancy when its total weight exactly equals the buoyant force acting on it.

For navigation and detection, the submarine relies on SONAR (Sound Navigation and Ranging). An "active" sonar system emits a high-frequency pulse of sound (a "ping") and listens for the echo. The time delay of the returning echo is used to calculate the distance to an object, and the Doppler shift of the echo reveals the object's relative velocity. The reactor core is a source of intense radiation and must be surrounded by heavy shielding, such as lead, to protect the crew.

Data:

Data for a typical nuclear submarine:

Table 1: Data for Nuclear Submarine

Parameter	**Value**
Total volume of the submarine's hull, V_{hull}	8.20×10^3 m^3
Mass of submarine with empty ballast tanks	7.40×10^6 kg
Density of seawater, ρ_{sea}	1030 kg m^{-3}
Thermal power output of nuclear reactor, P_{th}	160 MW
Efficiency of turbine/generator system, η	32%
Resistive drag force at 15 m s^{-1}, F_D	1.20 MN
Frequency of active sonar pulse, f	12.0 kHz
Speed of sound in seawater, v_{sound}	1520 m s^{-1}
Linear attenuation coefficient of lead for gamma rays, μ	57 m^{-1}

Questions

(a) The submarine is at the surface with its ballast tanks empty.

- (i) State Archimedes' principle.
- (ii) Calculate the total weight of the submarine.
- (iii) When fully submerged, calculate the maximum buoyant force on the submarine.
- (iv) Hence, determine the mass of seawater that must be pumped into the ballast tanks for the submarine to achieve neutral buoyancy (i.e., to float while fully submerged).

(b) The submarine's nuclear reactor powers its propulsion.

- (i) Calculate the total electrical power, P_{elec}, generated by the turbine system.
- (ii) The electric motors use this power to propel the submarine. Calculate the propulsive power required to overcome the drag force and maintain a constant speed of 15 m s^{-1}.
- (iii) Calculate the efficiency of the propulsion system (from electrical power to propulsive power).

(c) The submarine uses its active sonar system.

- (i) Calculate the wavelength of the 12.0 kHz sonar pulse in seawater.

(ii) The sonar emits a pulse which returns from the seabed 0.480 s later. Calculate the depth of the submarine.

(d) The sonar pulse reflects from another object that is moving **away** from the submarine.

(i) State how the frequency of the returning echo will differ from the emitted frequency.

(ii) What is the name of this phenomenon?

(e) The reactor is shielded to protect the crew from gamma radiation. The intensity I of radiation after passing through a thickness x of shielding is $I = I_0 e^{-\mu x}$.

(i) Calculate the half-value thickness (HVT) of lead.

(ii) Calculate the percentage of the original radiation intensity that remains after it passes through 15 cm of lead shielding.

Solutions

(a) (i) Archimedes' principle states that the upward buoyant force exerted on a body immersed in a fluid is equal to the weight of the fluid that the body displaces.

(ii) $W = mg = (7.40 \times 10^6) \times (9.81) = 7.26 \times 10^7$ N.

(iii) $F_{buoyant,max} =$ Weight of displaced water $= \rho_{sea} V_{hull} g$
$F_{buoyant,max} = (1030)(8.20 \times 10^3)(9.81) = 8.28 \times 10^7$ N.

(iv) For neutral buoyancy, $W_{total} = F_{buoyant,max}$.
$W_{total} = W_{sub} + W_{water} = 8.28 \times 10^7$ N.
$W_{water} = (8.28 \times 10^7) - (7.26 \times 10^7) = 1.02 \times 10^7$ N.
Mass of water $m_{water} = \frac{W_{water}}{g} = \frac{1.02 \times 10^7}{9.81} = 1.04 \times 10^6$ kg.

(b) (i) $P_{elec} = P_{th} \times \eta = (160 \times 10^6) \times 0.32 = 5.12 \times 10^7$ W (or 51.2 MW).

(ii) $P_{propulsive} = F_D \times v = (1.20 \times 10^6) \times (15) = 1.80 \times 10^7$ W (or 18.0 MW).

(iii) $\eta_{prop} = \frac{P_{propulsive}}{P_{elec}} = \frac{1.80 \times 10^7}{5.12 \times 10^7} = 0.352 = 35.2\%$.

(c) (i) $v = f\lambda \implies \lambda = \frac{v_{sound}}{f} = \frac{1520}{12.0 \times 10^3} = 0.127$ m.

(ii) The pulse travels to the seabed and back (distance $2d$).
$d = \frac{v_{sound} \times t}{2} = \frac{(1520)(0.480)}{2} = 365$ m.

(d) (i) The frequency of the returning echo will be **lower** than the emitted frequency.

(ii) The **Doppler effect**.

(e) (i) HVT is the thickness $x_{1/2}$ where $I = I_0/2$.
$x_{1/2} = \frac{\ln(2)}{\mu} = \frac{0.693}{57} = 0.0122$ m (or 1.22 cm).

(ii) $x = 15$ cm $= 0.15$ m.
$\frac{I}{I_0} = e^{-\mu x} = e^{-(57)(0.15)} = e^{-8.55} = 1.94 \times 10^{-4}$.
Percentage remaining $= (1.94 \times 10^{-4}) \times 100\% = 0.019\%$.

Synoptic Question 20 — Variant 2 (Higher Difficulty)

The Physics of a Nuclear Submarine: Extended Analysis

Instructions (CIE P4 style):

- Answer all questions.
- Show clear working for all calculations.
- Use an appropriate number of significant figures.
- For open-ended parts, credit is given for coherent physics reasoning.

Data for this variant

Parameter	Value
Hull volume, V_{hull}	$8.20 \times 10^3\,\text{m}^3$
Mass with empty ballast tanks, m_0	$7.40 \times 10^6\,\text{kg}$
Density of seawater, ρ	$1030\,\text{kg}\,\text{m}^{-3}$
Thermal reactor output, P_{th}	$160\,\text{MW}$
Turbine-generator efficiency, η_{tg}	32%
Drag at $15\,\text{m}\,\text{s}^{-1}$, F_D	$1.20\,\text{MN}$
Speed of sound in seawater, v_s	$1520\,\text{m}\,\text{s}^{-1}$
Sonar emission frequency, f	$12.0\,\text{kHz}$
Lead attenuation coefficient, μ	$57\,\text{m}^{-1}$
Take g	$9.81\,\text{m}\,\text{s}^{-2}$

Questions

(a) Buoyancy, stability and trim [12]

(i) Calculate the maximum upthrust on the submarine when fully submerged. [2]

(ii) Determine the ballast-water mass required for neutral buoyancy. [2]

(iii) If ballast water is accidentally shifted so that $1.5 \times 10^5\,\text{kg}$ moves $18\,\text{m}$ aft from the centre of mass, calculate the change in pitching moment about the centre of mass. [2]

(iv) *Open-ended:* Explain, using force and moment ideas, why poor trim control can increase energy consumption even when average speed is unchanged. [6]

(b) Power chain and endurance [13]

(i) Calculate electrical output power from the reactor-turbine system. [2]

(ii) Calculate the propulsive power needed to maintain $15\,\text{m}\,\text{s}^{-1}$. [2]

(iii) Hence find propulsion efficiency from electrical to propulsive output. [2]

(iv) The submarine now increases speed to $18\,\text{m}\,\text{s}^{-1}$. Assume drag force scales as v^2. Determine the new drag force and propulsive power. [3]

(v) *Open-ended:* Discuss two engineering limits (other than fuel availability) that may prevent sustained high-speed operation in a nuclear submarine. [4]

(c) Sonar, Doppler and tactical detection [13]

(i) Calculate sonar wavelength in seawater. [2]

(ii) A seabed echo returns after 0.480 s. Find depth. [2]

(iii) An echo from a target is received at 11 960 Hz when transmitted frequency is 12 000 Hz. Assuming a low-speed approximation for monostatic active sonar, use $\Delta f \approx \dfrac{2v_r}{v_s} f$ to estimate radial speed v_r of the target relative to submarine, and state direction of motion. [4]

(iv) *Open-ended:* Explain why temperature and salinity layering can create "shadow zones" for active sonar, and suggest one operational response. [5]

(d) Radiation shielding and design trade-offs [12]

(i) Derive the half-value thickness for lead from $I = I_0 e^{-\mu x}$ and evaluate numerically. [3]

(ii) Find percentage intensity remaining after 15 cm of lead. [2]

(iii) *Open-ended:* A designer proposes doubling shielding thickness around part of the reactor compartment. Evaluate this choice in terms of crew safety, mass, buoyancy/trim impact, and overall platform performance. [7]

Sample Solution

(a) Buoyancy, stability and trim

(i) $F_B = \rho V g = (1030)(8.20 \times 10^3)(9.81) = 8.29 \times 10^7\,\text{N}$.

(ii) Neutral buoyancy requires $m_{\text{total}} = \rho V = 8.446 \times 10^6\,\text{kg}$.
$m_{\text{ballast}} = m_{\text{total}} - m_0 = (8.446 - 7.40) \times 10^6 = 1.05 \times 10^6\,\text{kg}$.

(iii) Pitching moment change:
$\Delta M = \Delta m\, g\, d = (1.5 \times 10^5)(9.81)(18) = 2.65 \times 10^7\,\text{N m}$ (aft-down moment).

(iv) Poor trim changes angle of attack of hull/control surfaces, increasing hydrodynamic drag. More motor power is needed to maintain depth and heading corrections. Continuous control-surface deflection creates additional induced drag and turbulence. Reactor-to-propulsion chain then operates less efficiently because more output is diverted to overcoming avoidable resistive forces.

(b) Power chain and endurance

(i) $P_{\text{elec}} = \eta_{\text{tg}} P_{\text{th}} = 0.32(160 \times 10^6) = 5.12 \times 10^7\,\text{W}$.

(ii) $P_{\text{prop}} = F_D v = (1.20 \times 10^6)(15) = 1.80 \times 10^7\,\text{W}$.

(iii) $\eta_{\text{prop}} = \dfrac{P_{\text{prop}}}{P_{\text{elec}}} = \dfrac{1.80 \times 10^7}{5.12 \times 10^7} = 0.352 \approx 35.2\%$.

(iv) At $18\,\text{m s}^{-1}$, $F \propto v^2$:
$F_{18} = 1.20 \times 10^6 \left(\dfrac{18}{15}\right)^2 = 1.73 \times 10^6\,\text{N}$.
$P_{18} = F_{18} v = (1.73 \times 10^6)(18) = 3.11 \times 10^7\,\text{W}$.

(v) Limits include thermal rejection from steam/electrical systems and propulsion-cavitation/acoustic penalties. At high shaft power, propeller cavitation raises detectability and may damage blades. Mechanical stress/vibration and sustained reactor-side heat-flow constraints also cap safe continuous speed.

(c) Sonar, Doppler and tactical detection

(i) $\lambda = \dfrac{v_s}{f} = \dfrac{1520}{12000} = 1.27 \times 10^{-1}\,\mathrm{m}.$

(ii) $d = \dfrac{v_s t}{2} = \dfrac{(1520)(0.480)}{2} = 3.65 \times 10^{2}\,\mathrm{m}.$

(iii) $\Delta f = 12000 - 11960 = 40\,\mathrm{Hz}.$
$v_r \approx \dfrac{\Delta f\, v_s}{2f} = \dfrac{(40)(1520)}{2(12000)} = 2.53\,\mathrm{m\,s^{-1}}.$
Echo frequency is lower, so target is moving away (increasing range).

(iv) Layered temperature/salinity changes alter sound speed with depth, refracting sound rays away from some regions. This produces low-illumination "shadow zones" where echoes are weak or absent. Operational responses: change depth to a better acoustic layer, vary ping geometry/frequency, coordinate with passive sonar or another platform for multi-static coverage.

(d) Radiation shielding and design trade-offs

(i) For half-value thickness $x_{1/2}$, set $I/I_0 = 1/2$:
$\frac{1}{2} = e^{-\mu x_{1/2}} \Rightarrow x_{1/2} = \dfrac{\ln 2}{\mu} = \dfrac{0.693}{57} = 1.22 \times 10^{-2}\,\mathrm{m}.$

(ii) $x = 0.15\,\mathrm{m}$ gives
$\dfrac{I}{I_0} = e^{-57(0.15)} = e^{-8.55} = 1.94 \times 10^{-4},$
percentage $= 1.94 \times 10^{-2}\% \approx 0.019\%.$

(iii) Doubling shielding strongly reduces dose and improves crew safety margins, especially for long patrols. However added mass changes displacement margin, ballast requirement, and trim distribution. Increased mass may lower agility and increase propulsive demand; space and maintenance access may worsen. A balanced design targets ALARA dose principles while preserving buoyancy control, structural integrity, and mission performance.

Mark Scheme (CIE P4 style)

Total: 50 marks

Question part	Expected points for credit	Marks
(a)(i)	Correct use of $F_B = \rho V g$ and correct value (8.29×10^7 N)	2
(a)(ii)	Correct neutral mass idea; ballast mass $\approx 1.05 \times 10^6$ kg	2
(a)(iii)	Uses $\Delta M = \Delta m g d$ and correct magnitude/direction	2
(a)(iv)	Any 3 valid linked physics points on trim, control forces, drag, energy penalty (2 marks each)	6
(b)(i)	Correct electrical power from efficiency relation	2
(b)(ii)	Correct propulsive power from $P = Fv$	2
(b)(iii)	Correct efficiency ratio and percentage	2
(b)(iv)	Correct drag scaling and new power value (1+2)	3
(b)(v)	Two justified engineering constraints, with physics connection (2+2)	4
(c)(i)	Correct wavelength calculation	2
(c)(ii)	Correct time-of-flight halving and depth value	2
(c)(iii)	Correct Δf, correct Doppler estimate, correct direction	4
(c)(iv)	Explains refraction/layering + one feasible response; coherent physics chain	5
(d)(i)	Correct derivation and value of HVT	3
(d)(ii)	Correct substitution and final percentage	2
(d)(iii)	Balanced evaluation: safety gain, mass/buoyancy effects, performance trade-off, reasoned conclusion	7

Examiner guidance for open-ended parts: Award marks for valid, logically connected physics statements. Credit alternative correct reasoning. For high-level responses, expect clear cause–effect links and use of relevant terminology (e.g., drag, moment, cavitation, acoustic refraction, dose attenuation, buoyancy margin).